2. Continuous random variables

What students need to learn:

The concept of a continuous random variable.	
The probability density function and the cumulative distribution function for a continuous random variable.	Use of the probability density function $\mathrm{f}(x)$, where $\mathrm{P}(a < X \leqslant b) = \int_a^b \mathrm{f}(x)\,\mathrm{d}x$. Use of the cumulative distribution function $\mathrm{F}(x_0) = \mathrm{P}(X \leqslant x_0) = \int_{-\infty}^{x_0} \mathrm{f}(x)\,\mathrm{d}x$. The formulae used in defining $\mathrm{f}(x)$ will be restricted to simple polynomials which may be expressed piecewise.
Relationship between density and distribution functions.	$\mathrm{f}(x) = \dfrac{\mathrm{dF}(x)}{\mathrm{d}x}$.
Mean and variance of continuous random variables.	Knowledge and use of $\mathrm{E}(aX + b) = a\mathrm{E}(X) + b$ $\mathrm{Var}\,(aX + b) = a^2\,\mathrm{Var}\,(X)$
Mode, median and quartiles of continuous random variables.	